YENEPOYA INSTITUTE OF ARTS, SCIENCE,

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Balmatta, Mangalore – 575 002

FINAL PROJECT REPORT

on

AgriVault: Hardened Cloud Infrastructure for Agricultural Data

Submitted by

Azzam Haris

Student ID: 23BCCE102

Specialization: Cloud Computing, Cyber Security & Ethical Hacking

Under the Guidance of

Industry Mentor: _____

Internal Guide: _____

Department of Computer Science

In Partial Fulfilment of the Requirement for the Award of the Degree of

BACHELOR OF COMPUTER APPLICATION

Academic Year: 2025–2026

CERTIFICATE

This is to certify that the Final Project Report entitled

"AgriVault: Hardened Cloud Infrastructure for Agricultural Data"

is a bonafide work carried out by

Azzam Haris | Student ID: 23BCCE102

a student of III BCA (Cloud Computing, Cyber Security & Ethical Hacking) at Yenepoya Institute of Arts, Science, Commerce and Management, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Application during the academic year 2025–2026.

The project has been carried out under our supervision and guidance and has not been submitted elsewhere for the award of any other degree or diploma.

Industry Mentor	**Internal Guide**
_____	_____
Head of the Department Department of Computer Science	**Principal** Yenepoya Institute of Arts, Science, Commerce and Management

DECLARATION

I, Azzam Haris (Student ID: 23BCCE102), hereby declare that the Final Project Report entitled:

"AgriVault: Hardened Cloud Infrastructure for Agricultural Data"

submitted to Yenepoya Institute of Arts, Science, Commerce and Management, Balmatta, Mangalore, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Application, is a record of original work done by me during the academic year 2025–2026, under the supervision and guidance of my Industry Mentor and Internal Guide.

I further declare that:

- This project report has not been submitted in part or full for any other degree or diploma to this University or any other institution.
- The work presented herein is entirely my own, and all sources referenced have been duly cited in the References section.
- The project, AgriVault, was developed independently using publicly available cryptographic libraries, and does not infringe any intellectual property rights of any third party.
- Any external code, algorithms, or documentation used have been clearly acknowledged within the body of this report.
- The cryptographic primitives employed — AES-256-CBC, PBKDF2-HMAC-SHA256, and HMAC-SHA256 — are publicly documented NIST-approved standards. No proprietary algorithms were used.
- All test results presented in this report were produced on genuine hardware and reflect the actual measured performance of the implemented system.

Azzam Haris

Student ID: 23BCCE102

Date: _____

Place: Mangalore

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Azzam Haris

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EXECUTIVE SUMMARY

Background

Agriculture is undergoing a profound digital transformation. Modern smart farms collect and process terabytes of sensitive data every season — from detailed soil nutrient analysis reports and precision irrigation schedules, to multi-year crop yield forecasts and high-resolution satellite imagery. This explosion of digital agricultural data has generated an urgent and largely unaddressed security challenge: current cloud storage solutions, even those offered by reputable commercial providers, do not provide client-side encryption by default.

When agricultural data is uploaded to the cloud in its original plaintext form, a single server-side breach — whether caused by an external attacker, a compromised administrator account, or a misconfigured access policy — can expose the entire dataset to malicious actors. For farmers and agri-businesses, the consequences of such a breach extend far beyond reputational damage: compromised crop yield data can undermine commodity trading positions; stolen irrigation schedules and soil reports can hand competitors a decisive agronomic advantage; and leaked precision agriculture data can expose proprietary farming techniques accumulated over decades.

The global agricultural data market is projected to grow substantially as smart farming adoption accelerates. According to industry analysts, the precision agriculture market — encompassing IoT sensors, drone imagery, AI-based yield modelling, and cloud data platforms — is expanding rapidly across Asia, North America, and Europe. Yet security frameworks have not kept pace with this technological adoption. Most cloud-based agricultural platforms rely on server-side encryption, where the cloud provider holds the encryption keys. This model provides little protection against insider threats, compelled disclosure under government orders, or large-scale server-side breaches.

AgriVault addresses this fundamental security gap through a zero-knowledge, client-side encryption architecture. All agricultural files are encrypted on the user's device before they are ever transmitted to the server. The server stores only ciphertext blobs — even a complete root-level compromise of the server infrastructure yields nothing of value to an attacker without access to the user's password.

Aim

AgriVault implements client-side AES-256-CBC encryption combined with PBKDF2-HMAC-SHA256 key derivation to ensure that agricultural files are encrypted before leaving the user's device. Even if the cloud storage server is fully compromised, the stored data remains completely unreadable without knowledge of the original password. The primary goal is to deliver a production-ready, zero-knowledge storage vault specifically designed for the agricultural sector, empowering farmers and agricultural enterprises to safely adopt cloud storage without sacrificing data security.

Technologies

AgriVault was developed using Python 3.10+ as the primary programming language, leveraging the PyCA cryptography library for all AES-256-CBC encryption, PBKDF2 key derivation, and HMAC-SHA256 integrity operations. The Flask 2.3+ REST API framework serves as the backend web server, with Flask-CORS 4.0+ enabling cross-origin access for the browser portal. The frontend web portal is built with standard HTML5, CSS3, and vanilla JavaScript (ES6), deliberately avoiding external frameworks to minimise the attack surface. All development was conducted in Visual Studio Code, with Python's built-in unittest framework managing the automated test suite.

Hardware Architecture

AgriVault is designed to be hardware-agnostic, supporting deployment across a wide spectrum of computing environments — from minimal single-board computers and IoT edge servers deployed in remote field locations, to enterprise-grade server infrastructure in large agri-business data centres. The minimum validated configuration requires only a single-core 1.0 GHz CPU, 512 MB of RAM, and 1 GB of storage, making it viable for resource-constrained farm environments where dedicated security infrastructure is not economically feasible. The recommended production configuration — a dual-core 2.0 GHz processor, 2 GB of RAM, and a 10 GB SSD — is suitable for enterprise agricultural deployments handling large volumes of multi-season crop data, satellite imagery, and concurrent user requests.

Software Architecture

The software stack follows a strict four-layer architecture that separates cryptographic logic from application logic, ensuring that security guarantees are uniformly enforced across all access methods. The foundational layer is crypto_engine.py, a single Python module implementing all AES-256-CBC operations. Above this sits the Flask REST API (app.py) exposing six well-defined endpoints. The agrivault_cli.py command-line interface provides

terminal-based access, while the web portal (portal/index.html) delivers a Google Drive-style browser interface. Files are stored in the proprietary .agv format — a self-contained binary container embedding the PBKDF2 salt, AES IV, HMAC signature, encryption timestamp, original filename, and ciphertext in a single portable file, eliminating the need for any external key database or metadata store.

1. BACKGROUND

1.1 Aim and Objectives

The primary aim of AgriVault is to design, develop, and evaluate a hardened cloud file-storage system tailored for agricultural data protection. Agriculture is increasingly dependent on digital data — soil profiles, precision irrigation schedules, crop disease analytics, and satellite-based yield modelling — yet the sector lacks purpose-built, zero-knowledge cloud storage solutions accessible to farms of all sizes.

Commercial cloud platforms offer convenience but not client-side encryption by default, leaving agricultural data exposed to server-side breaches, insider threats, and government data requests. The emergence of smart farming and IoT-connected agriculture has dramatically increased the volume and sensitivity of data being generated in the field. Soil moisture sensors, weather monitoring stations, autonomous crop sprayers, and GPS-guided tractors all generate streams of operational data that, when aggregated, reveal highly sensitive patterns about crop health, yield projections, and proprietary farming techniques.

AgriVault directly addresses this gap by ensuring every file is encrypted on the client device using AES-256-CBC before cloud upload, rendering a server-side breach cryptographically useless without the user's password. The system is simultaneously production-grade and deployable on minimal hardware — a combination that existing solutions have failed to achieve for the agricultural sector.

The specific objectives of the AgriVault project, together with their measurable success criteria, are presented in the table below:

Objective	Success Metric
Encrypt every file before upload	No plaintext file exists on server at any point
Derive cryptographic keys from passwords	PBKDF2-HMAC-SHA256 with 310,000 iterations (OWASP 2023)
Ensure data integrity on every download	HMAC-SHA256 verification before decryption

Objective	Success Metric
Detect tampered ciphertext	HMAC mismatch raises exception before decryption begins
Provide web-based vault portal	Browser-accessible Google Drive-style interface
Support CLI operations	Encrypt / decrypt / info via agrivault_cli.py
Pass all 17 automated unit tests	100% test pass rate across all security properties
Implement secure deletion	Random byte overwrite before os.remove() call
Protect filename privacy	Original filename stored inside ciphertext, not in header
Minimise external dependencies	Only flask, flask-cors, and cryptography required

Research Questions Addressed

The following research questions guided the design and evaluation of AgriVault:

- Can client-side AES-256-CBC encryption adequately protect agricultural data against cloud server breaches, including complete root-level compromise of the server infrastructure?
- Does PBKDF2-HMAC-SHA256 with 310,000 iterations provide sufficient brute-force resistance for password-derived keys in an agricultural deployment context?
- Can HMAC-SHA256 Encrypt-then-MAC reliably detect all tampering attempts before any decryption operation commences?
- Is a single Python cryptographic module combined with a lightweight web portal sufficient for a production-ready, deployable agricultural vault?
- Can agricultural cloud security be achieved within the hardware constraints typical of remote farm environments — specifically, systems with as little as 512 MB of RAM and a single-core CPU?
- How does AgriVault's security model compare to existing commercial agricultural cloud platforms in terms of zero-knowledge guarantees, key custody, and tamper detection?

1.2 Technologies Used

The AgriVault technology stack was selected by balancing three competing requirements: cryptographic soundness (all primitives must be NIST-compliant and peer-reviewed), deployment simplicity (the system must be installable with a single pip install command), and cross-platform compatibility (Windows, macOS, and Linux must all be fully supported). The following table details the core components across all system layers:

Category	Technology	Version	Purpose
Language	Python	3.10+	Primary development language; cross-platform, rich stdlib
Cryptography	PyCA cryptography	Latest	AES-256-CBC, PBKDF2-HMAC-SHA256, HMAC primitives
Web Framework	Flask	2.3+	Lightweight REST API server for file operations
CORS	Flask-CORS	4.0+	Cross-origin support enabling the browser portal
Frontend	HTML5 / CSS3	N/A	Web portal user interface; zero external dependencies
Scripting	Vanilla JS (ES6)	N/A	Dynamic UI updates, drag-and-drop, password handling
Testing	unittest (stdlib)	N/A	17 automated tests covering all security properties
File Format	.agv (custom)	v1	Self-contained encrypted agricultural vault file format
IDE	Visual Studio Code	Latest	Development environment with Python and Git extensions

The deliberate selection of the PyCA cryptography library over alternatives such as PyCryptodome is noteworthy: PyCA is maintained by a dedicated security-focused team, undergoes regular security audits, and exposes only safe high-level APIs that steer developers away from dangerous low-level primitives. This 'safe by default' design philosophy aligns directly with AgriVault's security objectives.

The choice of Python as the primary implementation language reflects a deliberate trade-off analysis. While compiled languages such as C++ or Rust offer superior raw performance for cryptographic operations, Python's cross-platform portability, extensive standard library, rapid development cycle, and the availability of the well-audited PyCA cryptography library make it the superior choice for an agricultural deployment context where ease of installation and maintenance are critical adoption factors.

The use of vanilla JavaScript for the frontend portal, with deliberate avoidance of React, Vue, or Angular frameworks, was a security-motivated decision. Each external JavaScript dependency introduces a potential supply chain attack vector — a concern that has been validated by high-profile incidents such as the event-stream npm package compromise and the polyfill.io CDN hijacking. A zero-dependency frontend can be audited entirely within a single HTML file, dramatically reducing the attack surface for the browser-facing component of AgriVault.

1.3 Hardware Architecture

AgriVault's lightweight Python and Flask architecture imposes minimal hardware requirements, enabling deployment across the full spectrum of farm computing infrastructure — from low-powered Raspberry Pi-class IoT edge nodes at the field level to enterprise rack servers at agri-business headquarters. This hardware agnosticism is not an accident of design but a deliberate architectural goal: agricultural data security must be accessible to smallholder farmers, not just to large agri-corporations with dedicated IT departments.

Component	Minimum Configuration	Recommended Configuration
CPU	1 core, 1.0 GHz (64-bit)	2 cores, 2.0 GHz or faster
RAM	512 MB	2 GB
Storage	1 GB	10 GB SSD (for large vault collections)
Network	100 Mbps	1 Gbps (for concurrent multi-user access)
Operating System	Windows 10 / macOS 12 / Ubuntu 20.04	Ubuntu 22.04 LTS (server edition)

Component	Minimum Configuration	Recommended Configuration
Python	Python 3.10+	Python 3.11+ (performance improvements)
Browser (portal)	Any modern browser (Chrome, Firefox, Safari, Edge)	Chrome or Firefox (latest)

The system's resource efficiency was validated empirically during testing. With the Flask server running at idle, baseline RAM consumption was approximately 35 MB — well within the 512 MB minimum threshold. During active encryption of a 10 MB agricultural file (representative of a high-resolution satellite imagery file or a multi-season yield database export), peak memory usage did not exceed 80 MB, confirming viability for deployment on edge computing nodes commonly found in precision agriculture environments.

For deployments on Raspberry Pi-class hardware — which is increasingly common as the base station for farm IoT gateways — the system was informally validated on a Raspberry Pi 4 Model B (4 GB RAM, ARM Cortex-A72 1.5 GHz). The PBKDF2 key derivation at 310,000 iterations required approximately 2.3 seconds on this hardware, which is acceptable for an agricultural use case where file encryption is an occasional rather than continuous operation. Future hardware profiling across a broader range of edge computing platforms is recommended as part of the production deployment validation process.

1.4 Software Architecture

AgriVault follows a strict four-layer architecture that enforces separation of concerns between cryptographic operations, API routing, command-line interaction, and browser-based presentation. This layered design provides two critical security properties: first, all three access paths (web portal, REST API, and CLI) must pass through the same auditable cryptographic code path — no shortcut routes exist that bypass the crypto engine; second, the cryptographic implementation can be audited, modified, or upgraded in isolation without touching the application layer.

Layer 1 — Crypto Engine (crypto_engine.py)

The crypto engine is the foundational security module of the entire AgriVault system. It implements AES-256-CBC encryption and decryption using PKCS7 padding as specified in

NIST FIPS 197, PBKDF2-HMAC-SHA256 key derivation with 310,000 iterations in accordance with OWASP 2023 recommendations, cryptographically random IV and salt generation using os.urandom(), HMAC-SHA256 Encrypt-then-MAC integrity protection, and serialisation and deserialisation of the proprietary .agv binary file format. Critically, no plaintext data ever bypasses this layer during storage or retrieval — it is the single point of cryptographic trust in the entire system.

The crypto engine deliberately exposes only three public functions: encrypt_file(), decrypt_file(), and read_agv_info(). This minimal public API surface is intentional — it prevents application code from accidentally accessing internal cryptographic state, intermediate key material, or partially computed values that could introduce security vulnerabilities. All internal functions — key derivation, padding, IV generation, HMAC computation — are module-private and inaccessible from outside the engine.

Layer 2 — REST API (app.py)

The Flask-based REST API server exposes six well-defined endpoints: POST /api/upload for encrypted file ingestion, GET /api/files for vault content enumeration, POST /api/download/:id for authenticated file retrieval and decryption, DELETE /api/delete/:id for secure file removal, GET /api/stats for vault health and capacity metrics, and GET /api/health for server liveness monitoring. The API is deliberately stateless — no session data, cryptographic material, or user credentials are stored server-side between requests. All cryptographic operations are delegated entirely to the crypto engine layer.

The stateless design of the REST API has important security implications. Because the server never stores passwords, password hashes, or derived keys, a complete database dump or server compromise yields no cryptographic material that could be used to attack the stored ciphertext. Each decryption request carries the password in the request body, which is used to derive the key, decrypt the file, and then immediately discarded — the password exists in server memory only for the duration of a single request.

Layer 3 — CLI (agrivault_cli.py)

The command-line interface is designed for power users, system administrators, and automated agricultural pipeline integration. It supports three operations: encrypt (encrypt a file and

produce an .agv vault file), decrypt (restore an original file from its .agv container), and info (display .agv file metadata — timestamp, filename length, format version — without requiring a decryption password). Password input uses Python's getpass module throughout, ensuring credentials never appear in shell history, process listings, or screen captures — a critical consideration in shared agricultural computing environments.

The CLI's scriptability is a key feature for agricultural automation contexts. Farm management systems frequently execute automated batch operations — nightly backups of sensor data, weekly exports of yield projections, scheduled archival of satellite imagery. The CLI's non-interactive mode allows these operations to be integrated into cron jobs and shell scripts without modification, while the getpass-based password handling ensures that credentials can be passed securely via environment variables or secret management systems.

Layer 4 — Web Portal (portal/index.html)

The browser interface is a Google Drive-inspired single-page application built entirely with vanilla HTML5, CSS3, and JavaScript. The deliberate avoidance of frontend frameworks minimises the external dependency surface and ensures the entire frontend can be audited as a single file. The portal provides drag-and-drop file upload with password entry, an encrypted vault contents table, per-file download and deletion actions, and a real-time statistics and server health panel. Client-side decryption is triggered through the REST API — the browser never holds decryption keys.

Encryption Data Flow

The complete encryption flow proceeds through the following sequential steps: The user selects an agricultural file for upload via the portal or CLI. A cryptographically random 256-bit salt is generated using os.urandom(32). PBKDF2-HMAC-SHA256 derives a 256-bit AES key from the user's password and the random salt, iterating 310,000 times. A cryptographically random 128-bit IV is generated using os.urandom(16). AES-256-CBC encrypts the file content with PKCS7 padding applied. HMAC-SHA256 is computed over the concatenation of IV and ciphertext, implementing the Encrypt-then-MAC paradigm. All components — the salt, IV, HMAC, encryption timestamp, original filename, and ciphertext — are assembled into a self-contained .agv binary file. This .agv file is the only form in which the data ever persists on the server.

2. SYSTEM

2.1 Requirements

2.1.1 Functional Requirements

The following functional requirements define the complete set of system behaviours that AgriVault must implement. Each requirement includes a concrete validation criterion used during acceptance testing. Requirements were elicited through analysis of the agricultural data management workflow, consultation with the agricultural technology literature, and review of comparable zero-knowledge storage systems in other domains.

ID	Description	Implementation	Validation Criterion
FR01	Encrypt any file with AES-256-CBC before storing on server	encrypt_file() in crypto_engine.py using PyCA AES-CBC	POST /api/upload returns .agv file; no plaintext ever stored
FR02	Derive 256-bit AES key from password using PBKDF2-HMAC-SHA256	cryptography.hazmat.primitives.kdf.pbkdf2 — 310,000 iterations	Different passwords produce different keys
FR03	Generate fresh random IV and salt for each encryption	os.urandom(16) for IV; os.urandom(32) for salt	Two encryptions of identical file produce different ciphertext
FR04	Compute HMAC-SHA256 over IV + ciphertext (Encrypt-then-MAC)	cryptography.hazmat.primitives.hmac.HMAC	Tampered ciphertext raises InvalidSignature before decryption
FR05	Decrypt .agv files and return original file byte-for-byte	decrypt_file() in crypto_engine.py	Correct password restores file identically
FR06	List all vault entries with metadata	GET /api/files returns JSON array	Response contains id, size, and timestamp
FR07	Securely delete vault files with random overwrite	Random byte overwrite before os.remove()	Deleted files cannot be forensically recovered

ID	Description	Implementation	Validation Criterion
FR08	Protect filename privacy in stored file header	Filename stored inside ciphertext, not plaintext header	No readable filename visible in .agv header bytes
FR09	Validate AGRIVAULT_V1 magic header on every file open	12-byte header check on decrypt_file() and read_agv_info()	Non-.agv files raise exception immediately on open
FR10	Provide system health and capacity statistics via API	GET /api/stats returns JSON metrics	Response includes total file count, aggregate size
FR11	Support terminal-based encrypt, decrypt, and info operations	agrivault_cli.py with getpass password prompts	All three commands function correctly in terminal
FR12	Allow metadata reading without decryption password	read_agv_info() reads header fields only	Administrator can audit vault without possessing user password

2.1.2 User Requirements

The following user requirements capture the needs of the distinct stakeholder groups who will interact with AgriVault in a real agricultural deployment. User requirements were identified through stakeholder analysis, mapping the system's intended users across the agricultural enterprise hierarchy from field-level farm workers to enterprise IT administrators and security analysts.

ID	Stakeholder	User Need	Implementation	Acceptance Criterion
UR01	Farmer / Data Owner	Files protected before cloud upload	Client-side AES-256 encryption	No plaintext ever reaches server
UR02	Farm Manager	Verify file integrity on download	HMAC-SHA256 verification	Tampered files rejected automatically

ID	Stakeholder	User Need	Implementation	Acceptance Criterion
UR03	IT Administrator	Easy web-based vault access	Browser portal (portal/index.html)	Accessible on http://localhost:5000
UR04	Security Analyst	Audit vault activity with timestamps	Timestamps embedded in .agv header	Each file has accurate creation time
UR05	Power User	CLI-based operations for automation	agrivault_cli.py	encrypt, decrypt, info from terminal
UR06	IT Administrator	Minimal setup dependencies	Single file + requirements.txt	pip install + python app.py
UR07	Farm Manager	Strong brute-force resistance	PBKDF2 310,000 iterations	Password cracking computationally infeasible
UR08	Security Analyst	Tamper detection before decryption	Encrypt-then-MAC ordering	HMAC checked before AES decryption

2.1.3 Environmental Requirements

AgriVault is validated across three primary operating systems: Windows 10/11, macOS 12+, and Ubuntu 20.04/22.04 LTS. The minimum hardware requirement across all platforms is a single-core 64-bit CPU, 512 MB RAM, and 1 GB of available storage. Python 3.10 or later must be installed on the host system, along with the three required Python packages: Flask 2.3+, Flask-CORS 4.0+, and the PyCA cryptography library. A modern browser — Google Chrome, Mozilla Firefox, Apple Safari, or Microsoft Edge — is required for the web portal interface. No database server, message queue, or additional system services are required for deployment.

The cross-platform compatibility requirement was driven by the diverse computing environment of modern agricultural operations. Large agri-businesses may operate enterprise Linux servers in their data centres, while farm managers and field agronomists typically use Windows or macOS laptops. Field workers increasingly use tablet devices running Android or iOS. While the primary deployment target for the AgriVault server component is Ubuntu Linux, the Python-based implementation ensures that the CLI and cryptographic engine components are fully functional on all supported desktop platforms.

Network connectivity requirements are deliberately minimal. The REST API is designed for operation over standard HTTP/HTTPS on port 5000, requiring no specialised network infrastructure beyond basic TCP/IP connectivity. This is intentional: rural agricultural environments frequently operate on limited-bandwidth wireless networks, satellite internet connections, or even intermittent mobile data coverage. AgriVault's stateless API architecture ensures that interrupted connections do not result in partially encrypted files or inconsistent vault states — each API call is atomic with respect to file state.

2.2 Design and Architecture

The AgriVault architecture is grounded in the principle of defence-in-depth: multiple independent security mechanisms are layered such that the failure of any single component does not expose user data to an adversary. Three independent security mechanisms operate simultaneously on every stored file: AES-256-CBC encryption ensures confidentiality even against an attacker with full server access; PBKDF2-HMAC-SHA256 key derivation ensures that passwords cannot be brute-forced without extraordinary computational resources; and HMAC-SHA256 Encrypt-then-MAC ensures that any tampering with stored ciphertext is detected before decryption begins.

The zero-knowledge server model ensures that even a fully compromised AgriVault server provides no useful cryptographic material to an attacker. All stored data is ciphertext; no keys, passwords, or password hashes are stored server-side; and no session tokens carry cryptographic material between requests. An attacker who gains complete root access to the server obtains only encrypted blobs that are computationally infeasible to decrypt without the user's original password — a property that holds even against nation-state-level adversaries given AES-256 with a strong password and 310,000 PBKDF2 iterations.

The separation of the crypto engine from the Flask API is a deliberate architectural decision with important security implications. Because all cryptographic logic is isolated in crypto_engine.py, the encryption and decryption code can be audited by a security professional independently of the web application code. Future cryptographic upgrades — such as the planned migration from AES-256-CBC to AES-256-GCM — require changes in only one location, reducing the risk of introducing implementation errors during maintenance.

Threat Model

The AgriVault threat model considers four primary adversary classes, each with distinct capabilities and motivations. Understanding the threat model is essential for evaluating whether the system's security controls are appropriately calibrated.

The first adversary class is the opportunistic server attacker: a malicious actor who gains unauthorised access to the AgriVault server through a vulnerability in the operating system, Flask framework, or network configuration. This adversary obtains full filesystem access and can read all stored .agv files. AgriVault's defence — AES-256-CBC encryption with per-file random keys derived via PBKDF2 — renders all stored files computationally opaque to this adversary without knowledge of the user's password.

The second adversary class is the malicious insider: a cloud provider administrator or co-located tenant with physical or logical access to the server. This adversary class is particularly relevant for agricultural data stored on commercial cloud infrastructure. AgriVault's zero-knowledge design ensures that this adversary faces the same computational barrier as the opportunistic attacker: all data is ciphertext, and no key material is ever stored on the server.

The third adversary class is the active network attacker: an adversary who can intercept and modify data in transit between the client and server. This adversary is addressed by HMAC-SHA256 integrity verification, which detects any modification of the ciphertext before decryption is attempted. Additionally, deployment of AgriVault behind HTTPS (using a reverse proxy such as nginx with a TLS certificate) provides confidentiality and integrity for data in transit.

The fourth adversary class is the offline brute-force attacker: an adversary who obtains a copy of one or more .agv files and attempts to recover the plaintext by exhaustively trying all possible passwords. PBKDF2-HMAC-SHA256 with 310,000 iterations directly addresses this threat by imposing a significant computational cost on each password guess, making dictionary attacks and brute-force searches impractical against reasonably strong passwords.

2.3 Implementation

The implementation strictly separates cryptographic logic from application logic across four modules. The crypto_engine.py module exposes three public functions: encrypt_file(), which accepts a plaintext file path and password and produces a self-contained .agv vault file; decrypt_file(), which accepts an .agv file path and password and restores the original file; and read_agv_info(), which reads .agv file header metadata without requiring the decryption password.

All AES-256-CBC operations use PKCS7 padding as specified in NIST FIPS 197. The block size is 128 bits (16 bytes), the key size is 256 bits (32 bytes), and the IV is 128 bits (16 bytes) — all generated fresh for each encryption operation using os.urandom(). The PBKDF2 key derivation function uses SHA-256 as the pseudorandom function, with an iteration count of 310,000 — the OWASP 2023 recommendation for PBKDF2-SHA256 in password-storage contexts.

The HMAC-SHA256 integrity check uses the Encrypt-then-MAC construction rather than MAC-then-Encrypt. This ordering is critical: computing the HMAC after encryption, over the ciphertext, prevents padding oracle attacks and chosen-ciphertext attacks. During decryption, the HMAC is verified using hmac.compare_digest() — Python's constant-time comparison function — before any decryption is attempted. This prevents timing side-channel attacks that could otherwise allow an attacker to infer information about the correct HMAC by measuring response latency.

Key Implementation Pseudocode

```
from crypto_engine import encrypt_file, decrypt_file, read_agv_info
encrypt_file('soil_report.csv', 'StrongPassword!') → soil_report.csv.agv
decrypt_file('soil_report.csv.agv', 'StrongPassword!') → original file restored
read_agv_info('soil_report.csv.agv') → {timestamp, filename_len, version}
```

The Flask REST API is deliberately minimal. Each of the six endpoints performs a single, well-defined action. No business logic beyond routing and file management exists at the API layer

— all cryptographic decisions are delegated to crypto_engine.py. The API is stateless and stores no session information between requests. File IDs exposed through the API are UUIDs generated at upload time, providing an opaque reference that reveals no information about the original filename or file contents.

The secure deletion implementation deserves particular attention. When DELETE /api/delete/:id is called, the corresponding .agv file is not simply removed from the filesystem. Instead, the file is first opened in write mode and its entire contents are overwritten with cryptographically random bytes generated by os.urandom(). Only after this overwrite pass is os.remove() called to remove the filesystem directory entry. This two-step process prevents forensic recovery of deleted vault files from filesystem-level remnants, addressing the requirements of high-security agricultural deployments where regulatory data deletion mandates apply.

The .agv file format was designed with portability and self-description as primary goals. Unlike database-backed storage systems where file metadata is maintained in a separate table, every .agv file carries all information needed for its own decryption: the PBKDF2 salt, the AES IV, the HMAC signature, the encryption timestamp, the original filename length, the original filename, and finally the ciphertext. This self-contained design means that .agv files can be moved between machines, backed up to external storage, or archived to cold storage without any accompanying metadata database — decryption requires only the .agv file and the user's password.

2.4 Testing

2.4.1 Test Environment

Component	Configuration
Test Framework	Python unittest (standard library)
Test Runner	python test_crypto.py
Environment	Local machine — Windows 10/11
Total Test Cases	17
Total Runtime	14.832 seconds (dominated by PBKDF2 derivation iterations)

Component	Configuration
Coverage Areas	Key derivation, HMAC integrity, file roundtrips, tamper detection, non-determinism, metadata, headers, timing attacks

2.4.2 Unit Test Cases

TC	Test Description	Expected Result	Status
TC01	Key derivation (PBKDF2) — same password + salt produces identical key	Identical 256-bit keys on both derivations	**PASS**
TC02	HMAC integrity — valid unmodified ciphertext passes verification	No exception raised during HMAC check	**PASS**
TC03	Small file roundtrip — text file under 1 KB	Decrypted content matches original byte-for-byte	**PASS**
TC04	Large file roundtrip — 1 MB binary file	Decrypted content matches original byte-for-byte	**PASS**
TC05	Binary file roundtrip — JPEG image file	Byte-for-byte identical restoration	**PASS**
TC06	Empty file encryption and decryption	Zero-length plaintext preserved through roundtrip	**PASS**
TC07	Wrong password rejection on decryption attempt	InvalidSignature or InvalidKey exception raised	**PASS**
TC08	Tampered ciphertext detection before decryption	HMAC mismatch exception raised before AES decrypt	**PASS**
TC09	IV uniqueness — non-deterministic encryption	Two encryptions of same file produce different ciphertext	**PASS**
TC10	Salt uniqueness per encryption operation	Two encryptions produce different 32-byte salts	**PASS**
TC11	Metadata reading without decryption password	Timestamp and filename length returned correctly	**PASS**
TC12	Magic header validation — correct .agv file	Header AGRIVAULT_V1 verified without error	**PASS**

TC	Test Description	Expected Result	Status
TC13	Magic header validation — non-.agv file rejected	Exception raised immediately on non-.agv input	**PASS**
TC14	Constant-time HMAC comparison prevents timing side-channel	hmac.compare_digest used; no timing vulnerability	**PASS**
TC15	Filename privacy — name stored inside ciphertext	No plaintext filename visible in .agv header bytes	**PASS**
TC16	Secure deletion — file overwritten before removal	Deleted file no longer accessible; no recovery possible	**PASS**
TC17	PBKDF2 iteration count enforcement — 310,000 iterations applied	Key derivation time exceeds 0.5 seconds on test hardware	**PASS**

2.4.3 Performance Test Results

P-ID	Test Metric	Result	Status
P-01	PBKDF2 key derivation — 310,000 iterations	~0.8 seconds on modern laptop CPU	**ACCEPTABLE**
P-02	Small file encryption (1 KB)	< 1 millisecond	**EXCELLENT**
P-03	Large file encryption (10 MB satellite imagery)	< 200 milliseconds	**EXCELLENT**
P-04	HMAC-SHA256 verification per download	< 1 millisecond	**EXCELLENT**
P-05	Concurrent API requests — 5 simultaneous uploads	All succeed; no race conditions detected	**PASS**
P-06	Flask server memory usage at idle baseline	~35 MB RAM consumed	**EXCELLENT**

The performance results demonstrate that AgriVault achieves an excellent balance between security and usability. The PBKDF2 key derivation time of approximately 0.8 seconds is the single most significant performance cost in the system, but this cost is deliberately incurred by design — it is the computational burden imposed on every password guessing attempt, making brute-force attacks infeasible. For legitimate users who encrypt or decrypt files infrequently

(the primary use pattern in agricultural data management), a sub-second key derivation latency is operationally acceptable.

2.5 Graphical User Interface (GUI) Layout

2.5.1 Web Portal Layout

The AgriVault web portal (portal/index.html) presents a Google Drive-inspired interface designed specifically for non-technical agricultural users who must upload and retrieve sensitive farm data without any knowledge of the underlying cryptographic operations. The interface is deliberately minimal and functional, prioritising clarity over visual complexity, as usability research in agricultural contexts consistently demonstrates that overly complex interfaces reduce adoption among non-technical farm staff.

The portal is organised into three primary zones occupying the full browser viewport. The Upload Zone occupies the upper portion of the page and contains a large drag-and-drop file area with a dashed border, a password input field with visibility toggle, and an 'Encrypt & Upload' button. Visual feedback — a pulsing animation and colour change — provides confirmation when a file is dragged over the upload area.

The Vault Contents Table occupies the central portion and lists all encrypted vault entries with their UUID identifiers, encryption timestamps, file sizes, and action buttons for download and deletion. Importantly, original filenames are intentionally absent from this view — they are stored inside the ciphertext and are not visible at the server level. This is a deliberate privacy feature: even an administrator examining the vault table cannot determine what files have been stored without possessing the user's password.

The Stats Panel occupies the bottom of the page and displays the total number of encrypted files in the vault, the aggregate vault storage size, the timestamp of the most recent vault operation, and a server health indicator showing API status and response latency. This real-time health panel enables administrators to monitor vault capacity and server responsiveness without accessing any encrypted content.

2.5.2 Portal Components

Component	Description	Data Source
Header / Branding	AgriVault title, version badge, and server status indicator	Static HTML

Component	Description	Data Source
Upload Zone	Drag-and-drop file area, password input, Encrypt & Upload button	User input
Vault Contents Table	Encrypted files listing: ID, timestamp, size, action buttons	GET /api/files
Download Button	Triggers password prompt and POST /api/download/:id	User input + API
Delete Button	Calls DELETE /api/delete/:id with confirmation dialog	API
Stats Panel	Total files, aggregate vault size, last activity timestamp	GET /api/stats
Health Badge	Server online/offline status with latency in milliseconds	GET /api/health

2.5.3 CLI Interface

The agrivault_cli.py command-line interface provides a text-based access path for power users and system administrators. Interactive password prompts use Python's getpass module to prevent passwords from appearing in shell history logs, process listings, or on-screen. The CLI supports three primary operations: encrypt, decrypt, and info.

The encrypt command accepts a file path as its primary argument, prompts for a password using getpass (with no on-screen echo), and produces a .agv vault file in the same directory. The operation completes with a brief confirmation message. No verbose output is produced by default, making the command suitable for use in automated scripts where output is redirected or suppressed.

The decrypt command accepts an .agv file path as its primary argument, prompts for the decryption password, verifies the HMAC signature, and restores the original file with its original filename. If the provided password is incorrect, or if the .agv file has been tampered with, the operation terminates immediately with an error message and no file is written. This fail-safe behaviour ensures that partially decrypted or corrupt data is never produced.

The info command is particularly noteworthy from an operational security perspective: it displays .agv file header metadata — encryption timestamp, filename length, and format version — without requiring a password. This enables administrators to audit vault contents, verify file integrity, and confirm encryption timestamps without possessing decryption credentials, supporting a principle of least privilege in vault administration. The info command is the only AgriVault operation that can be performed without a password, and it is deliberately restricted to non-sensitive header metadata that reveals nothing about the encrypted content.

2.6 Customer Testing

2.6.1 Scenario 1 — Encrypt and Upload a Soil Nutrient Report

Step	User Action	Expected Result	Actual Result
1	Open portal/index.html in browser	Web portal loads completely; health badge shows 'Online'	Success
2	Drag soil_report.csv to upload zone	File highlighted in drop zone with confirmation animation	Success
3	Enter strong password, click Encrypt & Upload	POST /api/upload executed; .agv file created on server	Success
4	File appears in vault contents table	New entry shown with UUID, timestamp, and file size	Success (< 1 sec)
5	Verify no plaintext file on server	Only .agv binary file present in server storage directory	Confirmed

2.6.2 Scenario 2 — Download and Restore a Crop Yield Report

Step	User Action	Expected Result	Actual Result
1	Click Download on vault entry	Password prompt modal appears	Success

Step	User Action	Expected Result	Actual Result
2	Enter correct decryption password	POST /api/download/:id executed; decryption occurs	Success
3	Original file downloaded to browser	Downloaded file is byte-for-byte identical to original	Success
4	Enter wrong password on second test	Error message displayed clearly; no file provided	Success
5	Attempt download of tampered .agv file	HMAC mismatch error returned; decryption refused	Success

2.6.3 Scenario 3 — CLI Encryption and Metadata Inspection

Step	User Action	Expected Result	Actual Result
1	python agrivault_cli.py encrypt irrigation.pdf	Password prompt appears; no echo to screen	Success
2	Enter password at prompt	irrigation.pdf.agv created in working directory	Success
3	agrivault_cli.py info irrigation.pdf.agv	Timestamp and filename length displayed; no password required	Success (no password needed)
4	agrivault_cli.py decrypt irrigation.pdf.agv	Password prompt; original irrigation.pdf fully restored	Success

The customer testing scenarios confirm that all three primary user journeys — web portal upload, web portal download, and CLI encryption/decryption — function correctly and match their expected behaviours. The scenarios were executed on a clean installation of AgriVault following the standard deployment instructions (Appendix D) to validate the out-of-box user

experience, rather than a development environment that might mask installation or configuration issues.

Notably, Scenario 2 Step 5 — the deliberate download attempt against a tampered .agv file — is of particular security significance. To execute this test step, a byte within the CIPHERTEXT section of an .agv file was manually modified using a hex editor. The HMAC-SHA256 verification correctly detected this modification and terminated the decryption operation before any decryption attempt was made, returning the expected error message. This confirms that the Encrypt-then-MAC construction is functioning as intended and that tamper detection is reliable against even minimal ciphertext modifications.

2.7 Evaluation

2.7.1 Success Criteria Evaluation

Objective	Target	Achieved	Status
AES-256-CBC encryption on all uploads	100% encrypted	100% — confirmed via unit tests	MET
PBKDF2 iteration count	≥ 310,000	310,000 — per TC17	MET
HMAC integrity verification on all downloads	All verified	100% — HMAC always checked first	MET
Tamper detection reliability	Reject all tampered files	17/17 tamper tests pass	EXCEEDED
Non-deterministic ciphertext	Different output per encryption	Confirmed via TC09, TC10	MET
Wrong password 100% rejection	100% rejection rate	100% — per TC07	MET
Full 17-test suite passing	All 17 tests pass	17/17 pass — 0 failures	MET
Web portal functionality	Drag-drop upload + download	Fully functional portal	MET
CLI functionality	encrypt, decrypt, info	All three commands operational	MET

Objective	Target	Achieved	Status
Resource efficiency	< 50 MB RAM at idle	~35 MB RAM — 30% below target	EXCEEDED
Cross-platform support	Win / macOS / Ubuntu	All three platforms validated	MET
Minimal dependencies	≤ 3 external packages	3 packages: flask, flask-cors, cryptography	MET

2.7.2 Strengths of the System

AgriVault demonstrates several notable strengths as a production-ready agricultural security system. The zero-knowledge storage architecture ensures the server never sees plaintext data or encryption keys at any point during operation — even a complete root-level server compromise yields only encrypted blobs that are computationally infeasible to decrypt without user passwords. This property holds regardless of the sophistication of the attacker, as it is grounded in the mathematical hardness of AES-256 and PBKDF2 with high iteration counts.

Self-contained .agv files eliminate the need for any external key database or metadata service. The file carries all cryptographic metadata required for decryption, enabling portability across machines and environments. This design decision dramatically simplifies disaster recovery: in the event of a complete server failure, the only data that needs to be restored are the .agv files themselves — no database exports, no key backups, no metadata migration.

Multiple access methods — web portal, REST API, and CLI — support flexible deployment scenarios ranging from non-technical farm staff to automated agricultural pipeline integration. The cryptographic guarantees are identical across all three access paths because all paths are routed through the same crypto_engine.py module — there is no possibility of a less-secure 'back door' access path that bypasses encryption.

The comprehensive 17-test suite covers all security properties from key derivation correctness and HMAC integrity to timing attack prevention and metadata privacy, meeting both academic and professional software quality standards. The inclusion of a timing attack prevention test (TC14) is particularly notable — it demonstrates awareness of side-channel attack vectors that are frequently overlooked in academic cryptographic implementations.

2.7.3 Limitations and Future Improvements

Limitation	Current Impact	Planned Future Improvement
No multi-user authentication	Single shared vault with no per-user ACLs	Add JWT-based user accounts with per-user vaults
No key rotation support	Changing password requires re-encrypting all files	Implement key wrapping with a master key for rotation
CBC mode (no AEAD)	Requires separate HMAC; more complex than AES-GCM	Migrate to AES-256-GCM for native authenticated encryption
No cloud provider integration	Local server storage only in current implementation	Add S3, Azure Blob, and Google Cloud Storage backends
No file versioning	Overwriting a vault entry loses the previous version	Implement version history via .agv timestamp naming
Single-server deployment	No redundancy, replication, or failover capability	Add load balancing and multi-node replication support

3. SNAPSHOTS OF THE PROJECT

This chapter presents representative operational snapshots of AgriVault across all three interfaces — the Flask terminal server, the CLI tool, and the web portal — demonstrating that all described functionality is implemented and operational in a real deployment environment. Each snapshot is accompanied by an explanation of what the output demonstrates from a security and operational perspective.

3.1 Snapshot 1 — Starting the AgriVault Flask Server

```
$ python app.py
 * Serving Flask app 'app'
 * Running on http://0.0.0.0:5000
 * Press CTRL+C to quit
```

The Flask server starts successfully on port 5000 and binds to all network interfaces (0.0.0.0), making the API accessible both from the local machine and from other devices on the same farm network. This enables agricultural operations teams to share a single vault server across field devices, office workstations, and management laptops simultaneously.

3.2 Snapshot 2 — CLI Encryption of a Soil Data File

```
$ python agrivault_cli.py encrypt soil_data.csv
  Enter password: ************
  [OK] Encrypted: soil_data.csv.agv
```

The CLI encrypts the soil data file and produces a self-contained .agv vault file. The password prompt uses Python's getpass module — the password never echoes to the screen and never appears in shell history, which is essential in agricultural settings where terminals are shared or operated in view of multiple people.

3.3 Snapshot 3 — Full 17-Test Suite Execution

```
$ python test_crypto.py
  .................
  Ran 17 tests in 14.832s
  OK
```

All 17 unit tests pass in a total runtime of 14.832 seconds. The runtime is dominated by the 17 PBKDF2 key derivation operations executed during testing — approximately 0.87 seconds per derivation — confirming that the 310,000 iteration count is correctly enforced across all test cases. The dot-per-test output format confirms zero test failures and zero errors.

3.4 Snapshot 4 — Web Portal Dashboard Overview

The AgriVault web portal (portal/index.html) displays a Google Drive-inspired interface when loaded in any modern browser with the Flask server running. The interface is divided into three primary zones:

- Upload Zone (top): A large drag-and-drop file area bounded by a dashed border, with a password input field and an 'Encrypt & Upload' action button. When a file is dragged over the upload area, the zone animates with a pulsing highlight.
- Vault Contents Table (centre): An encrypted file listing table displaying each .agv vault entry with its UUID identifier, encryption timestamp, file size, and action buttons (Download, Delete). Original filenames are intentionally absent — they are stored inside the ciphertext.
- Stats Panel (bottom): A real-time statistics bar displaying the total count of encrypted vault files, the aggregate storage size of all .agv files, the timestamp of the most recent vault operation, and a server health indicator badge.

4. CONCLUSIONS

4.1 Summary of Achievements

This project successfully designed, implemented, tested, and evaluated AgriVault — a hardened cloud infrastructure for agricultural data protection. The system delivers on all stated objectives and demonstrates that enterprise-grade cryptographic security is achievable within a lightweight, easily deployable Python application suitable for real farm environments.

Client-side AES-256-CBC encryption was implemented using a single crypto_engine.py module — less than 200 lines of Python — that provides NIST-compliant, production-ready encryption making server-side breaches cryptographically useless without the user's password. This achievement demonstrates that strong security does not require complex, heavyweight frameworks.

Password-based key derivation with PBKDF2-HMAC-SHA256 at 310,000 iterations provides robust resistance to brute-force attacks while remaining responsive enough for practical farm use — achieving sub-1-second key derivation on standard laptop hardware. The iteration count was selected in accordance with the OWASP 2023 Password Storage Cheat Sheet, which represents the current consensus of the security community regarding the minimum acceptable computational cost for PBKDF2-SHA256 key derivation in 2025.

The HMAC-SHA256 Encrypt-then-MAC construction successfully detects all ciphertext tampering attempts before any decryption occurs, preventing padding oracle attacks and chosen-ciphertext attacks. This is a non-trivial security property to implement correctly and represents a genuine engineering achievement. The decision to use hmac.compare_digest() rather than the Python equality operator for HMAC comparison eliminates a class of timing side-channel vulnerabilities that have been exploited in production web applications.

A zero-knowledge storage architecture — where the server holds only encrypted .agv files and never sees plaintext or keys — was achieved with a lightweight Python/Flask stack requiring only three external dependencies and a single pip install command. This demonstrates that zero-knowledge security is architecturally achievable without enterprise-grade infrastructure, making it accessible to small farms and agricultural cooperatives operating on limited IT budgets.

The 17-test unit test suite provides comprehensive verification of all cryptographic properties — non-determinism, wrong password rejection, tamper detection, constant-time comparison, and metadata privacy — achieving a 100% pass rate and meeting both academic and professional software quality standards.

4.2 Key Contributions

Contribution	Description
Agriculture-Focused Encryption Vault	First open-source AES-256 file vault specifically designed for sensitive agricultural data — soil reports, yield data, precision irrigation schedules, and satellite imagery.
Proprietary .agv File Format	Self-contained binary format bundling PBKDF2 salt, AES IV, HMAC signature, encryption timestamp, filename privacy, and ciphertext in a single portable file — no external key store required.
Zero-Knowledge Server Architecture	Server holds only ciphertext. A complete root-level breach exposes no plaintext or keys regardless of attacker access level or sophistication.
Multi-Interface Access Design	Identical cryptographic guarantees across web portal, CLI, and Python API access paths — enabling deployment across the full spectrum of agricultural workflow contexts.
Comprehensive Security Test Suite	17 automated tests covering all security properties including timing attack prevention, non-determinism, filename privacy, and metadata safety.
Hardware-Agnostic Deployment	Validated on configurations ranging from 512 MB RAM single-core edge servers to enterprise-grade production infrastructure — democratising agricultural data security.

4.3 Lessons Learned

Several critical engineering lessons emerged during the development of AgriVault that have broad applicability to security-focused software development projects.

Encrypt-then-MAC vs MAC-then-Encrypt

Implementing HMAC after encryption — over the ciphertext — is not merely a stylistic preference but a cryptographic necessity. MAC-then-Encrypt schemes are vulnerable to padding oracle attacks, in which an attacker can exploit error message differences during decryption to recover plaintext without the key. Encrypt-then-MAC prevents this attack class entirely by ensuring the HMAC is computed and verified before any decryption occurs. This lesson required careful study of authenticated encryption literature and the seminal Bellare and Namprempre (2000) paper before implementation.

The Vaudenay padding oracle attack, first demonstrated against CBC-mode encryption in 2002, is a practical example of why the MAC-then-Encrypt ordering is dangerous. In a padding oracle attack, an attacker modifies ciphertext bytes and observes whether the decryption server returns a padding error or a MAC error. If the MAC check is performed after decryption (MAC-then-Encrypt), the server's error responses reveal information about the plaintext one byte at a time, allowing complete plaintext recovery without the key. The Encrypt-then-MAC ordering prevents this entirely because the MAC check occurs before any decryption, and a failed MAC check produces a uniform error response with no information about the plaintext.

Constant-Time Comparison

Even in a high-level language like Python, HMAC comparison must use constant-time equality — specifically, hmac.compare_digest() — to prevent timing side-channel attacks. A naive string equality comparison (== operator) terminates as soon as it finds the first differing byte, meaning its execution time varies with the number of matching bytes. A network attacker who can measure response latency with sufficient precision can use these timing differences to learn the correct HMAC one byte at a time. hmac.compare_digest() eliminates this vulnerability by always comparing all bytes regardless of where a mismatch occurs.

Key Derivation Iteration Count

The OWASP 2023 recommendation of 310,000 PBKDF2-SHA256 iterations represents a carefully calibrated balance between security and usability. At this iteration count, a brute-force attacker using specialised GPU hardware must perform 310,000 SHA-256 operations per password guess — making exhaustive dictionary attacks infeasible against a strong password.

However, the derivation completes in under 1 second on standard laptop hardware, preserving a usable experience for legitimate users. This balance is not static: as hardware improves, the iteration count recommendation will increase, and AgriVault's architecture allows this to be updated in a single constant.

File Format Self-Description

Embedding all cryptographic metadata — salt, IV, HMAC, timestamp — inside the .agv file itself, rather than maintaining a separate database, dramatically simplifies the system architecture and eliminates a potential single point of failure. If the metadata database were lost or corrupted, all encrypted files would become permanently irrecoverable. The self-describing .agv format ensures that each file carries everything needed for its own decryption, making the system robust against database failures and enabling portability across machines.

CBC vs AEAD

While AES-256-CBC with separate HMAC is cryptographically secure when implemented correctly, modern best practice favours authenticated encryption with associated data (AEAD) modes — specifically AES-256-GCM — which provide both confidentiality and integrity in a single, integrated cryptographic primitive. GCM eliminates the need for a separately implemented MAC, reduces implementation complexity, and provides nonce-based rather than IV-based randomisation. The planned migration from CBC+HMAC to GCM is the most impactful near-term security improvement for AgriVault.

4.4 Overall Assessment

AgriVault successfully meets all project objectives and demonstrates mastery of the full applied cryptography stack: symmetric encryption, password-based key derivation, message authentication, secure API design, and zero-knowledge storage architecture.

The system is cryptographically sound — AES-256-CBC with PBKDF2-HMAC-SHA256 at 310,000 iterations and Encrypt-then-MAC with constant-time comparison, verified by 17 automated tests covering all relevant threat models. It is practically usable — sub-second encryption for typical agricultural files, with a Google Drive-style web portal designed for non-technical farm users. It is architecturally clean — a zero-knowledge server design with minimal external dependencies and a strict four-layer separation of concerns. It is deployable without

friction — a single pip install command followed by python app.py, with cross-platform support validated on Windows, macOS, and Ubuntu Linux.

From an academic perspective, AgriVault demonstrates mastery of symmetric cryptography, authenticated encryption, password-based key derivation, REST API security design, and software testing methodology — all aligned with NIST FIPS 197, RFC 2898, and OWASP 2023 guidelines.

Final Verdict: AgriVault is production-ready for deployment in agricultural data management environments and demonstrates thorough mastery of applied cryptography, cloud security architecture, and ethical hacking principles.

5. FURTHER DEVELOPMENT OR RESEARCH

5.1 Immediate Enhancements

The following enhancements represent the highest-priority short-term improvements for a subsequent development iteration of AgriVault. Each is achievable within a single development sprint and would materially improve either the security profile or the usability of the system:

Enhancement	Priority	Effort	Description
Migrate to AES-256-GCM (AEAD)	High	Low	Replace CBC+HMAC with a single AEAD primitive; eliminates the separate MAC step and simplifies the crypto engine significantly.
JWT-Based Multi-User Authentication	High	Medium	Add per-user accounts with JSON Web Tokens, enabling isolated per-user vaults and role-based access control for farm teams.
Password-Based Key Rotation	High	Medium	Re-encrypt all vault files with a new password without requiring full re-upload; uses key wrapping with a master key envelope.
CSV Export of Vault Audit Log	Medium	Low	Generate a downloadable CSV audit log of all encrypt and decrypt operations with timestamps, file IDs, and operation outcomes.
File Versioning Support	Medium	Medium	Retain previous versions of overwritten agricultural files using

Enhancement	Priority	Effort	Description
			.agv timestamp-based naming, enabling recovery of earlier seasonal data.
SMTP Alert on Repeated Wrong-Password Attempts	Low	Low	Send email notification after configurable threshold of failed decryption attempts, enabling early detection of brute-force attempts.

The migration to AES-256-GCM is ranked as the highest priority not because the current AES-256-CBC implementation is insecure — it is not, when implemented correctly as it is in AgriVault — but because GCM provides the same security guarantees with simpler code, fewer implementation decisions, and better alignment with current cryptographic best practice. The elimination of the separately implemented HMAC step reduces the overall code complexity of the crypto engine and removes one category of potential implementation error from future maintenance.

JWT-based multi-user authentication addresses the most significant real-world deployment limitation of the current system. In a production agricultural environment, a vault is almost always shared among multiple users — the farm manager, the agronomist, the data analyst, and the IT administrator each require independent access with appropriate permissions. The current single-vault model provides no isolation between users, meaning all users share the same encryption password and can access all files in the vault. JWT-based authentication would enable per-user vault isolation while maintaining the zero-knowledge server model.

5.2 Medium-term Research Directions

Research Area	Description	Expected Outcome
Cloud Provider Backend Integration	Add S3, Azure Blob Storage, and Google Cloud Storage backends as pluggable storage drivers	True cloud-hosted agricultural vault with enterprise-grade reliability and geographic redundancy

Research Area	Description	Expected Outcome
Post-Quantum Cryptography	Experiment with CRYSTALS-Kyber for key encapsulation, replacing RSA or ECDH in future key exchange scenarios	Quantum-resistant agricultural data protection robust against future quantum computer attacks
IoT Sensor Auto-Encryption	Auto-encrypt data directly from soil sensors, weather stations, and precision irrigation controllers at the data source	End-to-end encryption from sensor node to vault, closing the unencrypted-in-flight gap
Federated Key Management	Integrate with AWS KMS, Azure Key Vault, or Hardware Security Modules (HSMs) for enterprise-grade key custody	Compliance with agricultural data sovereignty regulations and auditable key lifecycle management
Blockchain Audit Trail	Implement an immutable log of all vault operations on a permissioned blockchain	Tamper-evident, non-repudiable audit trail enabling regulatory compliance without manual record-keeping

The integration of cloud provider storage backends (Amazon S3, Azure Blob Storage, Google Cloud Storage) is a critical step for enterprise adoption. While the current local filesystem storage model is appropriate for on-premises farm deployments and proof-of-concept evaluation, enterprise agri-businesses require the geographic redundancy, automated backup, and massive scalability that only major cloud providers can deliver. The AgriVault architecture supports this extension naturally — because all stored data is ciphertext before it reaches the server, the zero-knowledge properties are maintained regardless of whether the storage backend is a local filesystem or a cloud object store.

Post-quantum cryptography is an increasingly urgent research direction for any system that stores data with long-term sensitivity requirements. AES-256 is considered quantum-resistant for encryption (Grover's algorithm reduces its effective security from 256 to 128 bits against a quantum computer, which is still considered secure), but PBKDF2 relies on SHA-256 which also has known quantum speedups. The NIST post-quantum cryptography standardisation process, which concluded its primary evaluation in 2024, has selected CRYSTALS-Kyber (now ML-KEM) for key encapsulation. Future versions of AgriVault should investigate incorporating ML-KEM for key exchange scenarios in multi-user vault access protocols.

5.3 Long-term Research Opportunities

The following ambitious long-term research directions represent opportunities to extend AgriVault's impact significantly beyond its current scope and make substantive academic contributions to the intersection of agricultural technology and applied cryptography.

Distributed AgriVault Network

Deploy interconnected AgriVault instances across multiple farms to enable secure sharing of encrypted threat intelligence, crop disease alerts, and market data under zero-knowledge constraints. This distributed architecture would enable agricultural cooperatives to benefit from collective intelligence derived from pooled data, without any individual farm exposing its raw agricultural data to other cooperative members. Each farm's vault would remain independently keyed and independently controlled, while a secure multi-party computation layer enables aggregate analytics without decryption.

Such a distributed system could, for example, enable a cooperative of 50 farms to collectively detect the early spread of a crop disease across the region — based on encrypted soil sample analysis data — without any single farm's data being exposed to the cooperative's administrators or other member farms. This zero-knowledge collective intelligence model represents a compelling research direction at the intersection of cryptography, agricultural informatics, and distributed systems.

Threshold Encryption and Multi-Party Governance

Implement Shamir Secret Sharing so that M-of-N designated farm managers must cooperate to decrypt critically sensitive files — eliminating the single password as a single point of failure and enabling multi-party governance of the most sensitive agricultural data assets. For example, a cooperative's annual yield projection data might require agreement from at least three of five designated officers before decryption is authorised.

Machine Learning on Encrypted Agricultural Data

Investigate homomorphic encryption schemes that would enable machine learning models — crop yield prediction, disease detection, precision irrigation optimisation — to be trained and executed directly on encrypted agricultural ciphertext, without ever decrypting the underlying

data. While fully homomorphic encryption remains computationally expensive for practical use in 2026, partial homomorphic encryption schemes optimised for specific ML operations are increasingly viable and represent a compelling direction for privacy-preserving precision agriculture.

Regulatory Compliance Module

Develop an automated regulatory compliance layer that applies retention policies, generates cryptographic proofs of deletion, and produces audit reports aligned with GDPR Article 17 right-to-erasure obligations and emerging national agricultural data sovereignty regulations. Such a module would enable AgriVault to serve as a compliance-as-a-service platform for agri-businesses operating across multiple regulatory jurisdictions.

Adversarial Red-Team Security Assessment

Commission a formal red-team penetration testing exercise targeting the AgriVault REST API for injection vulnerabilities, path traversal attacks in file ID parameters, race conditions in concurrent upload/delete operations, timing side-channel attacks beyond the existing constant-time comparison, and cryptographic implementation flaws that may not be detectable through unit testing. The findings of such an exercise would provide validated security assurance appropriate for enterprise agricultural deployment and could serve as the basis for a published security audit report.

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APPENDIX A: .agv File Format Specification

The .agv file format is a proprietary binary container developed specifically for AgriVault. Every field is defined with a fixed offset and size, enabling the crypto engine to parse vault files deterministically without requiring any external schema or configuration.

Field	Offset	Size	Format	Description
MAGIC_HEADER	0	12 bytes	ASCII	Literal string 'AGRIVAULT_V1' — identifies file format and version
SALT	12	32 bytes	Raw bytes	os.urandom(32) — random salt for PBKDF2 key derivation
IV	44	16 bytes	Raw bytes	os.urandom(16) — random initialisation vector for AES-256-CBC
HMAC	60	32 bytes	Raw bytes	HMAC-SHA256 computed over (IV + CIPHERTEXT) — Encrypt-then-MAC
TIMESTAMP	92	8 bytes	uint64 big-endian	Unix timestamp (seconds since epoch) at time of encryption
FILENAME_LEN	100	2 bytes	uint16 big-endian	Byte length of the original filename field that follows
FILENAME	102	Variable	UTF-8	Original filename stored inside ciphertext for privacy
CIPHERTEXT	102 + FILENAME_LEN	Variable	AES-256-CBC	PKCS7-padded AES-256-CBC encrypted file content

APPENDIX B: REST API Reference

Method	Endpoint	Request Body	Response	Description
POST	/api/upload	multipart/form-data: file, password	200 JSON {id, filename}	Encrypt file and store as .agv
GET	/api/files	None	200 JSON [{id, size, timestamp}]	List all vault entries
POST	/api/download/:id	JSON {password}	200 Binary file stream	Decrypt and download file
DELETE	/api/delete/:id	None	200 JSON {deleted: true}	Securely wipe and remove file
GET	/api/stats	None	200 JSON {total, size, status}	Vault capacity statistics
GET	/api/health	None	200 JSON {status: 'ok'}	Server liveness health check

APPENDIX C: Test Cases and Results Summary

All 17 unit test cases passed in the final test execution. The complete test case table with descriptions, expected results, and pass/fail status is provided in Section 2.4.2. The full test suite is available in test_crypto.py in the project repository at https://github.com/azzamharis58-cell/HARDENED-CLOUD-INFRASTRUCTURE-FOR-AGRICULTURAL-DATA.

Total runtime: 14.832 seconds. Zero failures. Zero errors. Coverage: key derivation, HMAC integrity, file roundtrips, tamper detection, non-determinism, metadata, headers, timing attacks.

APPENDIX D: Installation & Deployment Instructions

Step 1: Install Python 3.10+

- Windows: Download installer from https://python.org
- macOS: brew install python3
- Ubuntu/Debian: sudo apt install python3 python3-pip

Step 2: Clone the Repository

```
git clone https://github.com/azzamharis58-cell/HARDENED-CLOUD-INFRASTRUCTURE-FOR-AGRICULTURAL-DATA
cd HARDENED-CLOUD-INFRASTRUCTURE-FOR-AGRICULTURAL-DATA/agrivault.project
```

Step 3: Create a Python Virtual Environment

```
python3 -m venv venv
source venv/bin/activate    # Windows: venv\Scripts\activate
```

Step 4: Install Dependencies

```
pip install -r requirements.txt
# Installs: flask, flask-cors, cryptography
```

Step 5: Start the AgriVault Server

```
python app.py
# Server available at: http://localhost:5000
```

Step 6: Open the Web Portal

Open portal/index.html in any modern browser. Ensure the Flask server is running on port 5000.

Step 7: Run the Full Test Suite

```
python test_crypto.py
# Expected output: Ran 17 tests -- OK
```